

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 90,000 over 36 months is approved under the default policy (v1.0). Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,076 gives a FOIR of 14.2% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: LOW.

For the borrower:

Ravi Kumar, your loan of Rs 90,000 is approved. Your documents agree with each other, and your monthly instalment of Rs 3,076 fits within the affordability limit based on the income we verified.

Key drivers:

- FOIR 14.2% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 39,150 (volatility 4.1%)

Next steps:

- Issue sanction letter and collect e-mandate for the EMI.
- Archive the credit memo PDF with the file.

1. DECISION

Decision: APPROVED Confidence: 0.88

Reason codes: none

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is approved because every policy test passed.

Affordability and collateral are inside the limits: FOIR 14.2% vs limit 50.0% (-35.8 pp under); LTV 75.0% vs limit 80.0% (-5.0 pp under).

The model's risk band is LOW (score 0.2611 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.589038).

What pushes risk up: ltv (loan ÷ asset value) (75.0% vs a typical 72.4%, +0.005); repayment history score (70.0% vs a typical 70.5%, +0.001).

What pulls risk down: identity mismatch flag (no vs a typical yes, -0.150); foir (obligations ÷ income) (14.2% vs a typical 41.5%, -0.082); emi-to-income ratio (7.9% vs a typical 28.2%, -0.031).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 14.2% vs limit 50.0% (-35.8 pp under)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

PASS age 29 vs minimum 21 (+8 years)

PASS verified monthly income Rs 39,150

risk score 0.261123 = base 0.589038 + ? SHAP -0.327915 (method additive_exact, additivity error 0.0)

?(identity_mismatch_flag) = -0.1500 [value no vs baseline yes]

?(foir) = -0.0817 [value 14.2% vs baseline 41.5%]

?(emi_to_income) = -0.0305 [value 7.9% vs baseline 28.2%]

?(employment_stability) = -0.0262 [value 95.9% vs baseline 69.7%]

?(income_volatility) = -0.0210 [value 4.1% vs baseline 25.1%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-CLEAN-001

full_name: Ravi Kumar

age: 29

location_tier: tier_2

employment_type: gig_platform

declared_monthly_income: 39,000.00

loan_request.amount: 90,000.00

loan_request.down_payment: 30,000.00

loan_request.tenure_months: 36

loan_request.vehicle_price: 120,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Ravi Kumar conf 0.95 [application_form p.None] auto
monthly_income = 39,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 90,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 120,000.00 conf 0.95 [application_form p.None] auto
full_name = Ravi Kumar conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto
monthly_income = 38,900.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 2,500.00 conf 0.92 [bank_statement.txt p.1] auto
vehicle_price = 120,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 90,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 39,150.00
existing_obligations: 2,500.00
emi: 3,075.99
foir: 0.1424
ltv: 0.75
income_volatility: 0.0408
income_stability: 0.9592
loan_to_income: 0.1916
emi_to_income: 0.0786
input.annual_rate: 14.00
input.existing_monthly_obligations: 2,500.00
input.loan_amount: 90,000.00
input.monthly_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]
input.tenure_months: 36
input.vehicle_price: 120,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.2611 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 14% is comfortably serviceable
strengths: LTV 75% leaves borrower equity cushion
risks: INCOME_MISMATCH: Monthly income differs across documents by 0.3%
reasoning: Deterministic FOIR 14.24%, LTV 75.00%; XGBoost band LOW (score 0.2611).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.